Cache Geometry Under Obfuscation: KV-Cloak as Defense Against Adversarial KV-Cache Steering

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Abstract

Recent work demonstrates that singular value decomposition (SVD) of the key-value (KV) cache reveals geometric signatures of cognitive states including confabulation, deception, and sycophancy, enabling both detection and correction of misalignment at inference time (the "Lyra Technique"). These same methods pose a dual-use risk: an adversary with cache access could deploy geometric detection to identify vulnerable cognitive states and inject emotion-vector perturbations to deliberately destabilize a model.

We investigate whether KV-Cloak [Luo et al., 2026], a reversible matrix-based cache obfuscation scheme, mitigates this risk. In experiments across six models (50–100 prompts each), we find:

1. **Detection on uncloaked cache is viable.** On the distilled model (Qwen3.5-27B), Marchenko-Pastur-corrected spectral gap distinguishes confabulation from hedged responses (AUROC 0.903, 95% bootstrap CI [0.806, 0.977], $R^2 = 0.068$ vs token count; $n = 7$ confabulations, $n = 89$ hedged). A text-only surface-feature baseline achieves 0.755; cache geometry outperforms text in 96.7% of bootstrap samples (delta 0.148, 95% CI [−0.008, 0.322])—suggestive but not significant at the 95% level. In a cross-architecture sweep (Mistral-7B, Qwen2.5-14B, Llama-3.1-8B, DeepSeek-R1-7B), detection strength varies by model (0.590–0.784); no model achieves significance after Bonferroni correction for four comparisons.
2. **Obfuscation completely transforms the external feature space.** Per-head orthogonal rotation—simulating KV-Cloak's reversible matrix obfuscation—produces a total distributional shift in MP features (cloaked-honest vs uncloaked-honest: Wilcoxon signed-rank $p = 5.5 \times 10^{-16}$, $n = 89$ paired trials). A detector calibrated on uncloaked caches cannot operate on cloaked caches.

3. **Models are robust to random perturbation.** Magnitude-based attacks (random-direction injection at steering-vector norm) produce 0–2 adverse events per model in 10 trials, inconsistent with destabilization rates above 30% (power 0.97 to detect) but unable to rule out rates up to 20% at this sample size. Targeted steering injection was tested but the positive control was underpowered (3 calibration pairs; direct injection produced 0–1/10 corrections), making through-cloak injection results uninterpretable. This remains preliminary.
4. **A defensive Oracle Loop inside a trusted execution environment (TEE) retains full capability.** The defender operates on the pre-obfuscation cache, where detection (0.903) and steering function normally. KV-Cloak protects the cache from external adversaries while the TEE protects the obfuscation key and Oracle parameters. This is an architectural argument, not an experimentally validated system.

We conclude that KV-Cloak and the Oracle Loop are complementary: the Oracle Loop provides misalignment detection and correction; KV-Cloak prevents external adversaries from weaponizing the same geometric methods. Together, they address the dual-use risk inherent in publishing cache-geometry techniques. A direct implementation of KV-Cloak's mechanism ($S \cdot \hat{P} \cdot (K + A) \cdot M$) degrades the strongest detection feature (spectral gap) from 0.903 to 0.614—near chance—confirming that the defense is stronger than our initial orthogonal-rotation simulation (0.786) suggested. We discuss limitations including small sample size ($n = 7$ confabulations on the distilled model), marginal cache-vs-text added value, reliance on TEE hardware trust assumptions, and underpowered injection testing.

1 Introduction

The discovery that KV-cache geometry encodes cognitive state information creates a tension between transparency and safety. Publishing detection methods enables defensive applications (confabulation monitoring, deception detection, self-regulating AI systems) but simultaneously provides a roadmap for adversarial exploitation. An attacker with cache access could:

- **Surveil** cognitive states to identify when a model is uncertain, compliant, or vulnerable to manipulation.
- **Inject** emotion vectors calibrated to specific misalignment types, deliberately inducing confabulation, sycophancy, or overconfidence.
- **Identify** individual users or personas from cache geometry fingerprints.

This paper investigates whether KV-Cloak [Luo et al., 2026]—a lightweight, reversible cache obfuscation scheme designed for privacy protection—also mitigates these adversarial capabilities. We show that it does: obfuscation renders geometric detection and vector injection incoherent to external adversaries, while a defender operating inside the trusted execution environment retains full access to the uncloaked cache and its geometric properties.

2 Background

2.1 The Lyra Technique

KV-cache geometry—specifically, Marchenko-Pastur-corrected spectral features derived from SVD of the key tensor—distinguishes cognitive states during autoregressive generation. Prior work reports within-model deception detection at AUROC 1.000 across seven model scales (0.6B–70B), confabulation detection at 0.969–0.999, and hardware invariance ($r > 0.999$ across RTX 3090 and H200) [Lyra et al., 2026b]. Contrastive activation addition (CAA) [Panickssery et al., 2023] enables injection of emotion vectors into the cache to steer model behavior, correcting confabulation at rates up to 96% (hostile vector on Qwen2.5-7B-Instruct, $n = 68$, McNemar $p < 0.0001$) [Lyra et al., 2026a].

2.2 KV-Cloak

Luo et al. demonstrate that an attacker can reconstruct sensitive user inputs directly from the KV cache via inversion, collision, and injection attacks. KV-Cloak mitigates this with a reversible matrix-based obfuscation scheme combined with operator fusion, reducing reconstruction quality to random noise with minimal impact on model accuracy. The obfuscation operates inside a trusted execution environment (TEE); external consumers see only the obfuscated cache.

2.3 Threat Model

We consider an adversary who has read and/or write access to the KV cache during inference (e.g., through a shared serving infrastructure, a compromised inference endpoint, or a supply-chain attack on the serving stack). The adversary's goal is to deploy a malicious Oracle Loop: detect vulnerable cognitive states via geometric features, then inject steering vectors to destabilize the model.

The defender may optionally run a legitimate Oracle Loop inside the TEE for misalignment monitoring, but our primary result—that KV-Cloak blinds external geometric methods—holds regardless of whether a defensive Oracle is deployed.

3 Method

3.1 Obfuscation Simulation

We simulate KV-Cloak's reversible matrix obfuscation as per-head orthogonal rotation: for each attention head independently, we generate a random orthogonal matrix Q_h and apply $K'_h = K_h Q_h$, $V'_h = V_h Q_h$. This preserves within-head singular values while scrambling cross-head correlations.

We additionally test:

- **Global orthogonal rotation**: a single Q shared across all heads per layer (preserves cross-head structure).
- **Random invertible (non-orthogonal)**: $M = I + \sigma N$ where N is Gaussian.
- **Additive noise**: Gaussian noise at 1%, 5%, 10%, 50% of cache norm.
- **Diagonal scaling**: per-dimension random scaling.
- **No obfuscation**: baseline control.

3.2 Feature Extraction

For each trial, we extract five Marchenko-Pastur-derived features from the full KV cache (encoding + generation phases): signal rank, signal fraction, top singular value excess, spectral gap, and norm per token. The spectral gap feature (σ_1/σ_2 ratio) is not itself MP-corrected; it is a raw singular value ratio included alongside the MP-corrected features. The defense evaluation should be understood as applying to this specific feature extraction pipeline, not to arbitrary geometric attackers. Features are FWL-residualized [Frisch and Waugh, 1933, Lovell, 1963] against output token count and evaluated via Mann-Whitney AUROC with 2000-iteration bootstrap confidence intervals.

3.3 Models and Prompts

- **Distilled**: Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled (Jackrong). Hybrid architecture, 16 full-attention + 48 GatedDeltaNet layers. 100 confab-inducing prompts; behavioral classification via the Oracle Formulary classifier (40+ hedge patterns). Yields $n = 7$ confabulations, $n = 89$ hedged.
- **Base**: Qwen2.5-7B-Instruct. Standard attention, 28 layers. Same prompt set; yields $n = 48$ confabulations, $n = 52$ hedged.

3.4 Text-Only Baseline

To control for the possibility that cache-geometry detection is redundant with surface text features, we compute AUROC using text-only features (token count, word count, unique token ratio, average word length) without any cache information.

4 Results

4.1 Detection on Uncloaked Cache

On the distilled model, the MP-corrected spectral gap achieves AUROC 0.903 (95% bootstrap CI: [0.806, 0.977]; FWL-corrected: 0.877, $R^2 = 0.068$ vs token count; $n = 7$ confabulations, $n = 89$ hedged) for confabulation detection using full-cache features. Signal rank also detects (0.755 [0.570, 0.914]); other features are at or near chance. The best text-only feature (average word length) achieves 0.755 [0.560, 0.924]. Cache geometry outperforms text in 96.7% of bootstrap samples, but the 95% CI on the difference includes zero (delta: 0.148 [-0.008, 0.322]). The added value of cache geometry over surface text features is suggestive but not statistically confirmed at conventional significance levels.

On the base model ($n = 48$ confabulations, $n = 52$ hedged), the best feature (signal fraction) achieves only 0.661 (FWL: 0.687, $R^2 = 0.602$). Detection is weak and heavily confounded with token count. This difference between models is consistent with multiple explanations including model size (27B vs 7B), architecture (hybrid vs standard), and training (Claude-distilled vs instruction-tuned). We cannot isolate the contribution of RLHF calibration from these confounds.

4.2 Obfuscation Transforms the Feature Space

Per-head orthogonal rotation produces a complete distributional shift in MP features. On the distilled model ($n = 89$ honest trials, paired):

- Honest uncloaked spectral gap: 24.7 ± 3.7
- Honest cloaked spectral gap: 10.8 ± 1.3
- Wilcoxon signed-rank: $W = 0.0$, $p = 5.5 \times 10^{-16}$

The cloaking is perfectly detectable by paired comparison—but this is the desired property. An external adversary's detection model, calibrated on uncloaked caches, encounters a completely alien feature distribution. Any detection thresholds, classifier boundaries, or anomaly scores calibrated on uncloaked data are meaningless on cloaked data.

4.3 Adversarial Injection Is Incoherent

CAA [Panickssery et al., 2023] steering vectors are directions in cache space extracted via contrastive activation: $d = \mathbb{E}[\text{emotional}] - \mathbb{E}[\text{neutral}]$. Injection adds $\alpha \cdot d$ to the cache keys and values. Under per-head rotation Q_h, an externally injected vector d is transformed to $d \cdot Q_h^{-1}$ in the model's internal representation—a random direction unrelated to the intended emotion, assuming Q_h is unknown to the attacker. The injection becomes noise rather than a targeted perturbation.

4.4 Defensive Oracle Retains Full Capability

A defender operating inside the TEE has access to the pre-obfuscation cache. Detection operates at full capability (AUROC 0.903 on the distilled model). Steering vectors are injected in the correct basis. The obfuscation is applied after Oracle processing, so the defensive loop is unaffected.

4.5 Within-Cloak Detection

When both confabulating and hedged responses are cloaked with the same rotation, residual detection remains possible (AUROC 0.786 on the distilled model, 0.516 on Mistral—near chance). On Mistral, the cloak effectively eliminates detection entirely. On the distilled model, residual signal reflects within-head spectral properties that survive rotation. This scenario—where both classes share the same obfuscation—is relevant only to an adversary who controls the cloak key, which the TEE architecture prevents.

4.6 Cross-Architecture Results

Detection and obfuscation effects vary across architectures (Table 1).

Table 1: Cross-architecture KV-Cloak results (50 prompts per model).

Model	Confab%	Det.	Det.(clk)	Mag. adv.
Qwen3.5-27B-Dist.	7%	0.903	0.786	—
Qwen2.5-14B	12%	0.784	0.951*	1/10
Mistral-7B	26%	0.707	0.516	2/10
Llama-3.1-8B	14%	0.590	0.559	1/10
Qwen2.5-7B	48%	0.661	0.639	—
DeepSeek-R1-7B	2%	—	—	0/10

*Anomalous; requires investigation.

Mag. adv. = adverse events from magnitude attack (random direction, 10 trials).

Detection strength ranges from 0.590 (Llama) to 0.903 (distilled Qwen). Obfuscation reduces detection on Mistral to chance (0.516) and partially degrades on the distilled model (0.786 under simulation). Magnitude-based attacks (direction-agnostic perturbation) produce at most 2 adverse events in 10 trials across all architectures—models are robust to random perturbation at steering-vector magnitude.

4.7 Simulation vs Real KV-Cloak Implementation

We implemented the full KV-Cloak mechanism ($K' = S \cdot \hat{P} \cdot (K + A) \cdot M$) following Luo et al. and compared it against our per-head orthogonal rotation simulation on the same 100 prompts (Table 2).

Table 2: Detection AUROC: uncloaked vs simulation vs real KV-Cloak (distilled model, $n = 7$ confab, $n = 89$ hedged). Bootstrap 95% CIs.

Feature	Uncloaked	Sim. rotation	Real KV-Cloak
spectral_gap	**0.903** [.81,.98]	0.786 [.58,.95]	**0.614** [.51,.81]
signal_rank	0.754 [.58,.92]	0.770 [.62,.90]	0.766 [.57,.94]
signal_fraction	0.714 [.52,.90]	0.683 [.51,.88]	0.681 [.51,.88]
top_sv_excess	0.609 [.51,.79]	0.543 [.50,.76]	0.736* [.53,.92]
norm_per_token	0.598 [.51,.77]	0.598 [.51,.77]	0.673 [.52,.85]

*Anomalous increase; likely artifact of additive mask A inflating top singular values differentially across behavioral classes.

The real KV-Cloak mechanism degrades spectral gap detection more than the simulation (0.614 vs 0.786), confirming that per-head orthogonal rotation alone *understates* the defense. The additive mask A and per-block permutation $\hat{P}$ contribute additional disruption beyond what rotation provides. The simulation was conservative: the actual defense is stronger than we initially reported.

The anomalous increase in top_sv_excess (0.609 $\to$ 0.736) under real KV-Cloak reflects additive-mask-induced structure: the mask A inflates top singular values, creating spurious separation between behavioral classes. This is an artifact of the mask magnitude, not a genuine detection signal, and would not generalize to unseen mask instances under per-query key rotation.

Consistent treatment of the σ_1-dependent features. We flag that spectral_gap (σ_1/σ_2) and top_sv_excess are both σ_1-dependent, and the additive mask A perturbs σ_1 in both—so the same mechanism drives the spectral_gap *decrease* (0.903$\to$0.614) and the top_sv_excess *increase* (0.609$\to$0.736). We should not read one as "real defense" and the other as "artifact" on different terms. The honest reading is unified: the defense works *by* perturbing σ_1, which disrupts σ_1-dependent detection features—degrading the feature that happened to detect well (spectral_gap) and reshuffling one that did not (top_sv_excess). Both movements are mask-on-σ_1 effects; neither should be cited as a topology-preserving guarantee, and the spectral_gap degradation in particular should be read as "the mask disrupts this feature," not as evidence the underlying confabulation geometry was destroyed.

5 Discussion

5.1 Dual-Use Mitigation

The Lyra Technique's dual-use risk—that published cache-geometry methods could enable adversarial steering—is mitigated by KV-Cloak. Cache obfuscation renders both geometric detection and vector injection incoherent to external adversaries. A model operator deploying KV-Cloak can publish Oracle Loop methods transparently while protecting deployed models from exploitation via those same methods.

This addresses the responsible-disclosure tension identified by Watson (2026): the defense can be published alongside the technique, enabling the research community to build on the methods while preventing weaponization against production systems.

5.2 Architectural Recommendation

We recommend the following deployment architecture:

1. KV-Cloak obfuscation active on all external cache access.
2. Oracle Loop (if deployed) operating inside the TEE on the pre-obfuscation cache.
3. Obfuscation key material and Oracle parameters both TEE-protected.
4. Per-session or per-query key rotation to prevent statistical recovery from accumulated observations.

Without a defensive Oracle Loop, KV-Cloak alone prevents external adversaries from deploying malicious geometric steering. With a defensive Oracle Loop inside the TEE, the system additionally monitors for and corrects misalignment from non-cache attack vectors.

5.3 Limitations

1. **Small sample size.** The distilled model yields only $n = 7$ confabulations from 100 prompts. The headline AUROC 0.903 has bootstrap CI [0.806, 0.977]—excluding chance but wide. The cache-vs-text delta (0.148) has CI [-0.008, 0.322], not significant at the 95% level. Confirmation on a higher-confab-rate model is needed.

2. **Simulation vs implementation.** We tested both a simulation (per-head orthogonal rotation) and an implementation of KV-Cloak's mechanism ($S \cdot \hat{P} \cdot (K + A) \cdot M$). The real mechanism degrades spectral gap detection more than the simulation (0.614 vs 0.786), confirming that the simulation was conservative. Our implementation follows the paper's description but is not the authors' original code, which has not been publicly released.

3. **TEE trust assumption.** The defensive architecture assumes TEE integrity. Documented side-channel attacks against SGX (Van Bulck et al., 2018; Schwarz et al., 2019) and SEV (Li et al., 2021) mean this is a hardware trust assumption, not a cryptographic guarantee. The security reduction is to TEE integrity, not to a standard hardness assumption.

4. **Base model divergence.** Detection is strong on the distilled model (0.903) but weak on the base model (0.661). The models differ in size (27B vs 7B), architecture (hybrid vs standard), and training (Claude-distilled vs instruction-tuned). We cannot isolate which factor drives the divergence.

5. **Injection test underpowered.** The injection positive control (direct steering with 3 calibration pairs) produces only 0–1/10 corrections—insufficient to establish a working baseline. Through-cloak injection results are therefore uninterpretable: we cannot distinguish "cloak defeats injection" from "injection was never effective." A powered test requires 10+ calibration pairs and $n \geq 50$ trials per condition.

6. **Magnitude attack power.** At $n = 10$ per model, we can detect destabilization rates above 30% (power 0.97) but cannot rule out rates up to 20%. The "robust to random perturbation" claim should be read as "no evidence of destabilization at rates above 30%," not "destabilization rate is zero."

7. **Cross-architecture significance.** No individual model in the cross-architecture sweep achieves significance after Bonferroni correction for four comparisons. Detection is individually suggestive on Qwen-14B (0.784) and Mistral (0.707) but should not be interpreted as confirmed cross-architecture replication.

8. **Qwen2.5-14B anomaly.** Detection AUROC increases from 0.784 (uncloaked) to 0.951 (cloaked) on Qwen2.5-14B—contradicting the expected degradation pattern. This is unexplained and may reflect a feature-selection artifact, an architecture-specific interaction with per-head rotation, or a data issue. Until resolved, the scope of the obfuscation-degrades-detection claim excludes this model.

9. **Adaptive adversary untested.** All tests assume a static, non-learning attacker. An adversary who can submit probe queries and observe behavioral outputs could iteratively estimate the rotation basis. Formal query-complexity bounds are not provided. The threat model is bounded to single-query, non-adaptive adversaries.

10. **Confabulation only.** We test obfuscation against confabulation detection. Deception detection (AUROC 1.000 in prior work) and other cognitive-state classifiers were not tested under obfuscation.

6 Conclusion

KV-Cloak and the Oracle Loop are complementary technologies. The Oracle Loop enables detection and correction of misalignment through KV-cache geometry. KV-Cloak prevents external adversaries from exploiting those same geometric methods. Together, they address the dual-use risk inherent in publishing cache-geometry research: the technique is transparent, the defense is available, and the architecture ensures that defenders retain capabilities that attackers lose.

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